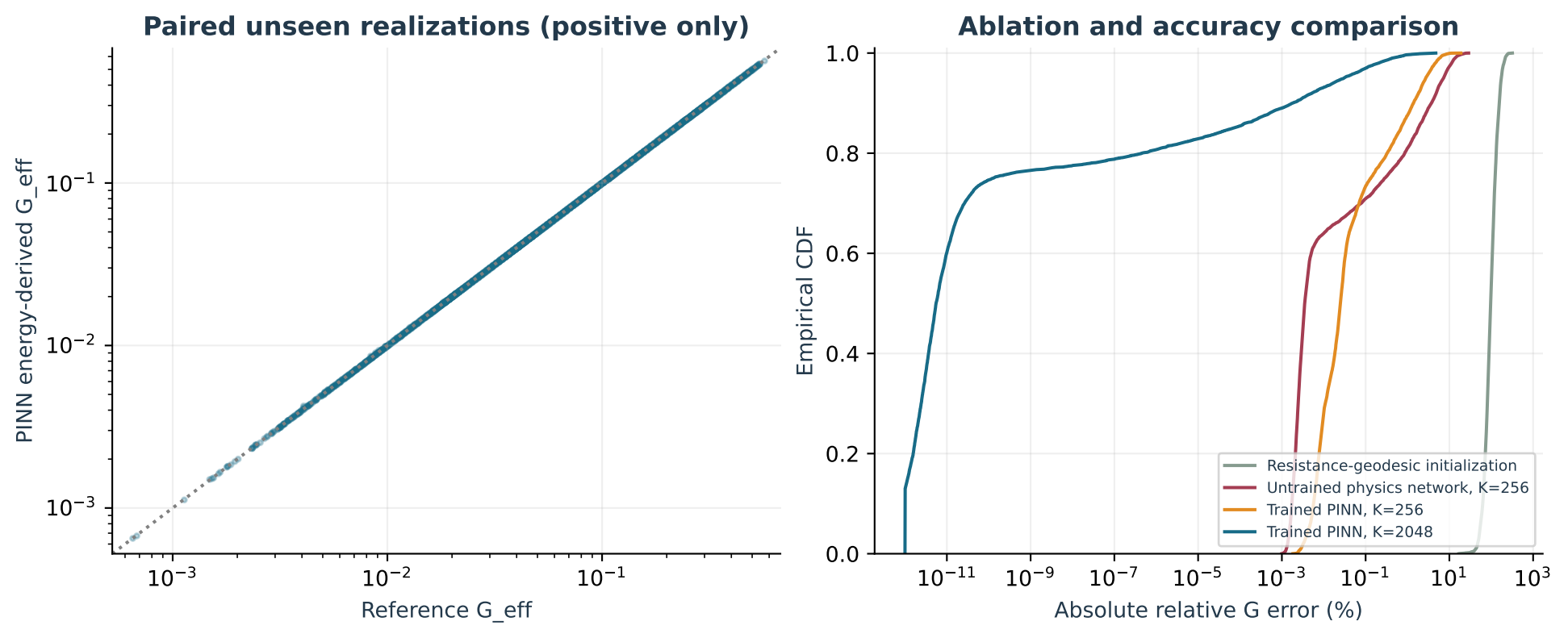
Paired unseen realizations (positive only)
PINN energy-derived G_eff
Reference G_eff
10^{-1}
10^{-2}
10^{-3}
10^{-3}
10^{-2}
10^{-1}
Ablation and accuracy comparison
Empirical CDF
Absolute relative G error (%)
1.0
0.8
0.6
0.4
0.2
0.0
10^{-11}
10^{-9}
10^{-7}
10^{-5}
10^{-3}
10^{-1}
10^{1}
10^{3}
Resistance-geodesic initialization
Untrained physics network, K=256
Trained PINN, K=256
Trained PINN, K=2048